

# Ganesh Ajjanagadde

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US citizen

## Education

Doctor of Philosophy (Ph.D.) in Computer Science at Massachusetts Institute of Technology (MIT) (2016-2020)  
GPA: 5.0/5.0  
Masters of Engineering (M.Eng.) in Electrical Engineering and Computer Science (EECS) at MIT (2015-2016)  
GPA: 5.0/5.0  
Undergraduate Bachelor of Science (SB) in EECS at MIT (2012-2015)  
GPA: 5.0/5.0  
Fellowship: Presidential Fellowship, NSF Graduate Fellowship, Claude E. Shannon Research Assistantship Award 2019  
Research Advisors: Professor Gregory Wornell, Professor Henry Cohn  
Research Areas of Interest: Signal Processing, Applied Mathematics, Coding Theory, Inference/Estimation/ML, Information Theory

## Work And Research Experience

*Research Scientist* Aug 2025 -  
Nvidia, Inc.

Research scientist in ADLR (applied deep learning research). Leading all sub 4-bit numerics technical effort at Nvidia (HW, SW, algorithms). On SW side, I wrote the fastest inference matmul kernels across industry, by incorporating novel SW defined numerical formats. They reach 1.77x speedup (98% of memory roofline) over CUTLASS NVFP4 baseline on Blackwell GPU at common inference batch sizes. On HW side, influenced Rubin and Feynman GPU generation sub 4-bit strategies and recommendations. Formats I codesigned for future GPU support sub 4-bit landscape at minimal SM area cost. On algorithm side, provided guidance to quantization aware distillation (QAD) teams for sub 4-bit formats. This effort achieved new precision Pareto frontier via QAD, beating common “2-bit is optimal point” folklore (e.g., ParetoQ paper).

Recognized internal expert on sub 4-bit numerics, fast kernel development, codesign, and algorithms.

*Senior Research Scientist/Software Engineer* Dec 2023 - Aug 2025  
Meta, Inc.

Senior research scientist in AI SW/HW codesign. Tech lead for numerical algorithms codesign. I led performance optimization work for Meta’s proprietary AI accelerator (MTIA), and also co-designed novel ML models/algorithms to make efficient use of HW (e.g., 2nd order optimizers for deep learning models that won AlgoPerf contest recently). I also made substantial contributions to future HW designs, saving x% of chip area, regarding efficient numerical formats (e.g., cascading GEMM recipes to emulate high precision that were independently discovered in academia as well: <https://arxiv.org/abs/2303.04353>), and through industry influence via vendor engagement regarding numerics. The cascading GEMM recipe I developed is incorporated in AMD MI355 SW stack for TF32 emulation. I also conceptualized and implemented proprietary math library for MTIA for high accuracy and great

performance, using MTIA HW features effectively. It is a complete FP32 math.h, with industry standard (CUDA) levels of accuracy. I was involved in both hands on work (training experiments, performance modeling scripts, ML kernel authoring, performant code prototypes, architecture analyses for HW, etc) as well as cross functional leadership and influence with involvement in road-mapping, goal setting, design documents, etc.

Recognized as an internal expert on low-precision numerics, quantization, and algorithms for high performance.

*Software Performance Architect*  
Apple, Inc.

July 2022 - Dec 2023

AI/ML power and performance engineer. Broadly responsible for making Siri workloads run fast while consuming minimal power. I brought a multi-faceted approach to the problem: all the way from the application layer by making ML model inference fast algorithmically to the system layer via good use of the underlying silicon and low level API's. Led critical power/perf efforts for Siri on Vision Pro, Watch, and HomePod. Recognized as AI/ML org expert on performance topics, and served as performance expert in Siri architecture working group.

Directly responsible for > 100% power improvements for Siri on Vision Pro enabling Siri to run on device, > 500 ms of Siri response time improvement on Watch via improvements ranging from kernel tunables to Siri natural language algorithmic improvements. On Watch this enabled Siri natural language on highly constrained (1.5 GB RAM) device with no latency regression.

Subsequently, worked on efficient inference of LLM on device, suitable for high powered Apple devices. Also worked on Siri on device for an even more constrained platform than Watch above (1 GB RAM HomePod Mini). Among other things, directly contributed to technical roadmap decision of on device Apple Intelligence (8 GB RAM iPhone, i.e iPhone 15 Pro+ only) via detailed technical analysis of the xnu kernel virtual memory system.

*Security/Privacy Software Engineer*  
Snap, Inc: remote

May 2020 - Jun 2022

Tech lead for application privacy at Snap, leading the efforts of 5 engineers. Deep expertise in both applied research and software engineering, with a passion and eye for high performance computation and optimization. Successful track record in combining these skills, learning new domains, and working cross-functionally to deliver great product experiences.

- Developed a novel, high performance, perceptual image hashing algorithm (similar to Apple NeuralHash) suitable for both clients and servers (e.g., hashes an image in less than 5 ms on common iPhones). Research underway to extend the algorithm to videos.
- Expanded the scope and delivered end to end encryption for Snap's AR glasses launched at Snap Partner Summit 2020. Worked on all aspects here: client (iOS, Android, C++) as well as server (Java backend).
- Co-owner of end to end encryption for messaging at Snap. Our system encrypts/decrypts  $\approx$  billion 1:1 snaps per day.
- Working on privacy preserving ad measurement technologies, especially relevant in view of iOS 14 changes. Implementation in modern C++ (C++17),

with bindings for other languages (Java, Go). In use today by some of Snap's partners.

- Performance geek - delivered a novel app binary size reduction, performance optimizations for our privacy preserving ad measurement technologies, deep optimizations for perceptual hashing etc. Extensive expertise with native (C/C++) code, both here as well as with my open source FFmpeg hacking in the past.
- Promoted at Snap for my work above.

*Graduate Student*

Jun 2015 - May 2020

Research Laboratory of Electronics, MIT, Cambridge, MA

Worked with Professor Henry Cohn on point configuration questions, primarily in Hamming space. These include packing and coding questions. Also worked under Professor Gregory Wornell on some computational imaging problems related to coded apertures, and with Professor Yury Polyanskiy on some classical questions of information theory, such as MAC (multiple access channel). Worked on statistical learning with a focus on classification problems for my Master's.

*Research Intern*

Jun-Aug 2019

Microsoft Research New England, Cambridge, MA

Worked with Prof. Henry Cohn on new lower bounds for the mean squared error of vector quantizers in high dimensions. We obtained the first rigorous improvements since the classical work of Zador, 1968.

*Research Intern*

Jun-Aug 2017

Analog Garage, Analog Devices

Worked on coming up with novel methods for anomaly detection in limited resource environments. Explored both classical learning as well as deep learning approaches.

*Draper Laboratory Undergraduate Research and Innovation Scholar* Sep 2014-May 2015

Laboratory for Information And Decision Systems, MIT, Cambridge, MA

Worked under Professor Yury Polyanskiy on problems of information theory.

*Software Engineer Intern*

Jun-Aug 2014

Kumu Networks, Santa Clara, CA

Wrote a hardware abstraction layer (HAL) for an RF circuit board.

**Links and Skills**

- GitHub: <https://github.com/gajjanag>
- Website: <https://gajjanag.github.io>
- Publications: <https://gajjanag.github.io/pubs.html>
- Languages: C++20, C99, Python (+numpy/scipy/matplotlib), CUDA, Julia, Java/Kotlin, Objective C, shell (bash/zsh), x86 asm, L<sup>A</sup>T<sub>E</sub>X, MATLAB, Verilog.
- Open source contributions: FFmpeg, Julia.
- Platforms/Tools: FFmpeg hacker, Linux, Android dev, git, Bazel, make, vim, GCP, PyTorch, Tensorflow/Keras, GLPK.